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Palaeomagnetic Studies in the British Caledonides–V Miscellaneous New Data

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Summary

Data are presented for a variety of Palaeozoic rocks from four areas in the British Isles: western Ireland (Arenig pillow lavas at Lough Nafooey, the Cambro–Ordovician Connemara Gabbros, Upper Silurian sediments of the Knocknaveen Group and Salrock Group, and intrusive rocks associated with the latter); the Ordovician of north Pembrokeshire (Trefgarn Andesitic Series, Fishguard Volcanic Series, and some basic intrusions); the Bristol District (Upper Llandovery lavas in the Tortworth Inlier, uppermost Devonian and basal Carboniferous red sediments); and the Orkney and Shetland Islands (middle Devonian to possibly Carboniferous lavas). These provide palaeomagnetic directions of limited reliability because of restricted sampling, outcrop, or stability evidence.

Three extensions of published studies are also reported. In the Hartshill Quartzite (Cambrian) the published result has not been substantiated. The contact aureole of the Arrochar Complex (Siluro–Devonian) provides a variety of stability evidence that its remanence dates from the time of intrusion of the igneous complex. The published results from Mweelrea ignimbrites (Llanvirn) are confirmed with minor modification.

Considered with other published data, these new results suggest that the pole relative to Britain remained in the same position from early Silurian to Lower (or even Upper) Devonian time, and comparatively rapid polar shift took place both before and after this interval.

1. Introduction

In this paper three reinvestigations of published results which could form cornerstones of any overall regional palaeomagnetic interpretation are described. One of these (Section 2) cannot be substantiated, but two (Sections 3 and 4) are reinforced by these new data. Some other studies have already made substantial additions to our knowledge, and they are briefly described here although they are not yet supported by the stability evidence of the standard attained in Papers I–IV of this series and by other definitive studies.

Also in the course of this survey a number of less satisfactory results were obtained; some show poor stability, and others are based on inadequate sampling

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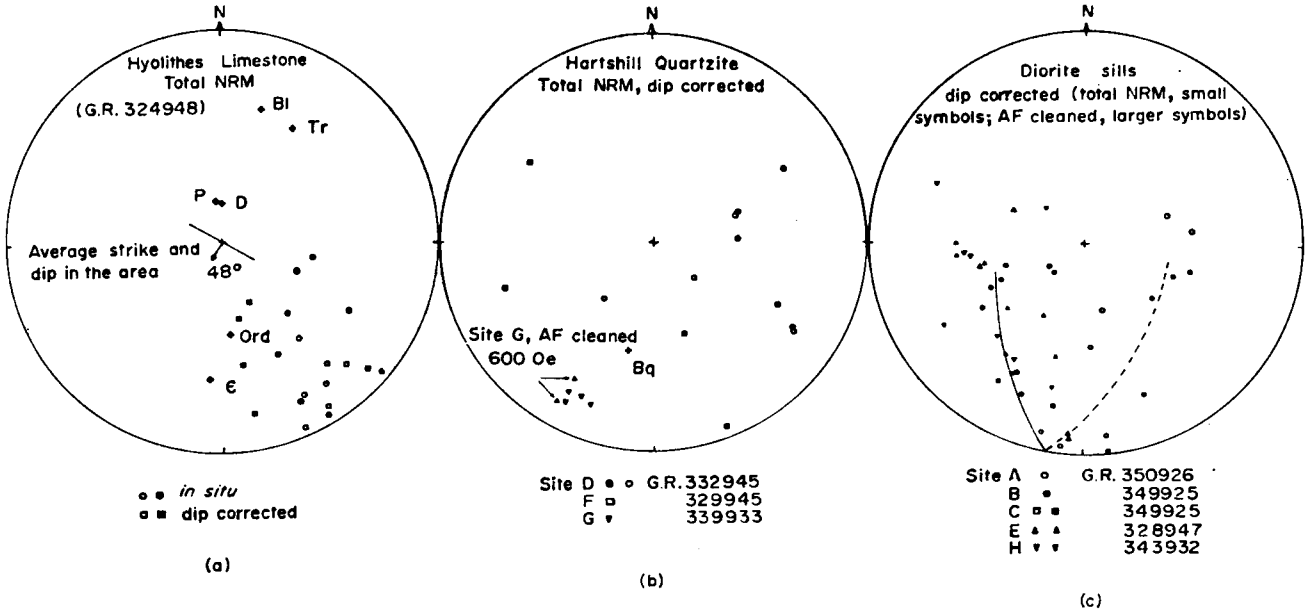


FIG. 1. Remanence directions in rocks from Nuneaton. In all stereographic projections in this paper solid symbols and crosses denote downward, positive inclination and open symbols negative inclination. P denotes the present local geomagnetic field, and D the geocentric axial dipole field. The Triassic field 'Tr' is inferred from studies on the New Red Sandstone in Britain (Clegg *et al.* 1954a, b; Creer 1957a, 1959); the Caerfai Series (Briden *et al.* 1971) can be interpreted as carrying remanence of either Cambrian or late Palaeozoic age and the corresponding inferred field at Nuneaton is denoted by 'ε'. An indication of the local Ordovician field is marked 'Ord' in the figure, being the mean of results in Paper I, and it is possible that this approximates the local Cambrian field also (see Paper VI, Section 2). The results of Bidgood (1961) on the limestones and quartzites are denoted by Bl and Bq.

which is either impossible to improve due to lack of outcrop or because it is not worthwhile to investigate further. These studies have not produced estimates of the palaeomagnetic field of sufficient precision or reliability to class them with the more definitive results (Table 1 of Paper VI), but nevertheless they all contribute to interpretations of the palaeomagnetic history of the region as a whole.

2. Cambrian rocks near Nuneaton

Bidgood (1961) published well grouped remanence directions from seven samples of Hartshill Quartzite after magnetic cleaning at 100 Oe. which are similar to commonly observed Palaeozoic field directions in Britain. He explained the different magnetization in the relatively porous interbedded *Hyolithes* Limestone as originating in Triassic time because its direction is similar to that found in New Red Sandstone rocks elsewhere in England. This interpretation is attractive because the New Red Sandstone overlies the Cambrian rocks in the sampling area, and reddening penetrates several feet downwards from the unconformity.

An attempt has been made to reinforce this study using strict stability criteria currently considered essential to an adequately defined palaeomagnetic result. One site was visited in the *Hyolithes* Limestone where it was reddened, and found to be magnetized irregularly to the south-east with inclination ranging from $-30°$ to $+50°$ *in situ* (Fig. 1(a)). Intensity in all five samples was about 3×10^{-6} Gauss with the remanence aligned close to the strike of the beds. This result is completely different from that of Bidgood (1961) and gives no hint of Triassic re-magnetization: south-easterly declinations in British rocks are typical only of the earliest magnetizations which can be shown to be original (i.e. Ordovician) and these particular samples of *Hyolithes* Limestone may possibly carry an original remanence; but because of the large scatter and absence of stability evidence little weight can be placed on this suggestion.

Four samples were taken at each of three sites in the Hartshill Quartzite (Fig. 1(b)). Two of these sites were weakly magnetized ($<0{\cdot}8 \times 10^{-6}$ Gauss except for two samples) and their remanent directions were widely scattered, in marked contrast to the results of Bidgood. At site G remanence was stronger ($\sim 6{\cdot}0 \times 10^{-6}$ Gauss), directions in all four samples agree closely, and there is some evidence of stability in alternating fields (AF) up to 600 Oe, after which the mean direction is $D = 206°$, $I = +10°$ (*cf.* Bidgood's 193°, +34°). Although it is likely that this remanence dates from the Palaeozoic, the data available from the Hartshill Quartzite, even incorporating Bidgood's results, are not sufficient to justify quoting a Cambrian pole position on their basis.

Five sites sampled in camptonite sills intruding Cambrian rocks at Nuneaton, also failed to produce clear cut results. The directions show a planar distribution (Fig. 1(c)) but the simplest explanation in terms of two components of ancient magnetization lying in this plane finds no support from the AF demagnetization tests carried out in fields up to 600 Oe.

3. Contact aureole of the Arrochar Intrusive Complex

A palaeomagnetic survey of the Arrochar Complex and its aureole was described by Briden (1970) who noted the general decrease of NRM intensity with increasing distance from the intrusive contact, and the differing character of AF demagnetization curves for samples closer to and further from the contact. Further work confirms the shape of the NRM profile (Fig. 2). In general the initial susceptibility (χ) profile of these same samples (Fig. 2) follows the intensity-profile. An upward 'spike' near 10 m, and a downward spike at 52 m in both curves are superimposed on a decreasing trend. Thus the effect of thermal metamorphism seems to be to increase both

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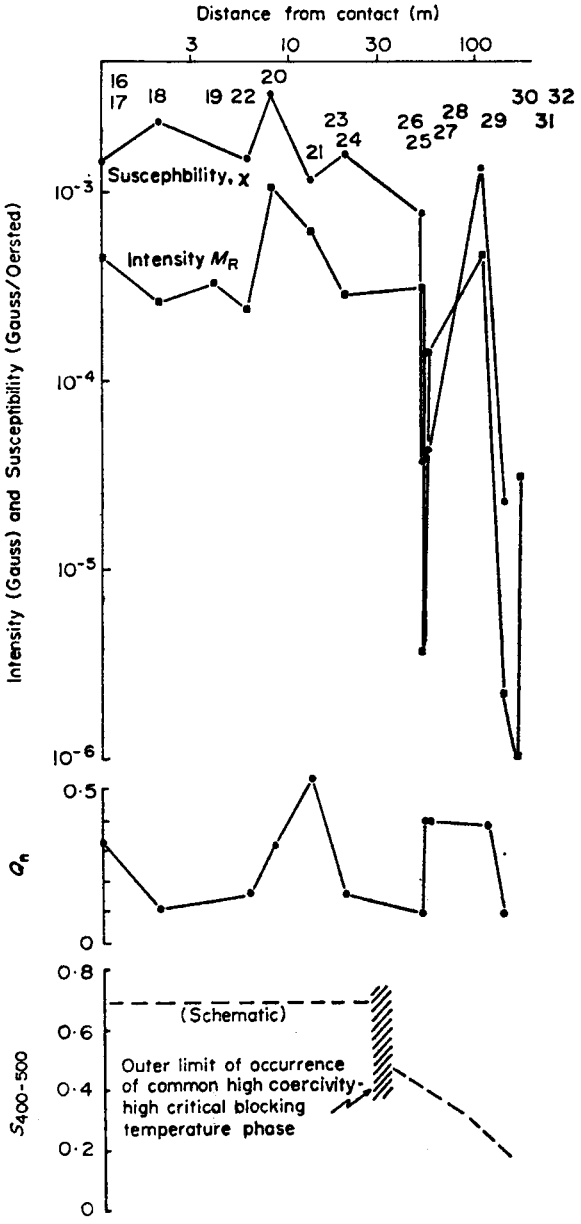


FIG. 2. Variation of total NRM (M_R), initial susceptibility (χ), Koenigsberger Ratio Q_n ($= M_R/\chi$), and generalized variation of SI in the range 400–500 Oe, with distance from the igneous contact with the Arrochar Complex at site L of Briden (1970).

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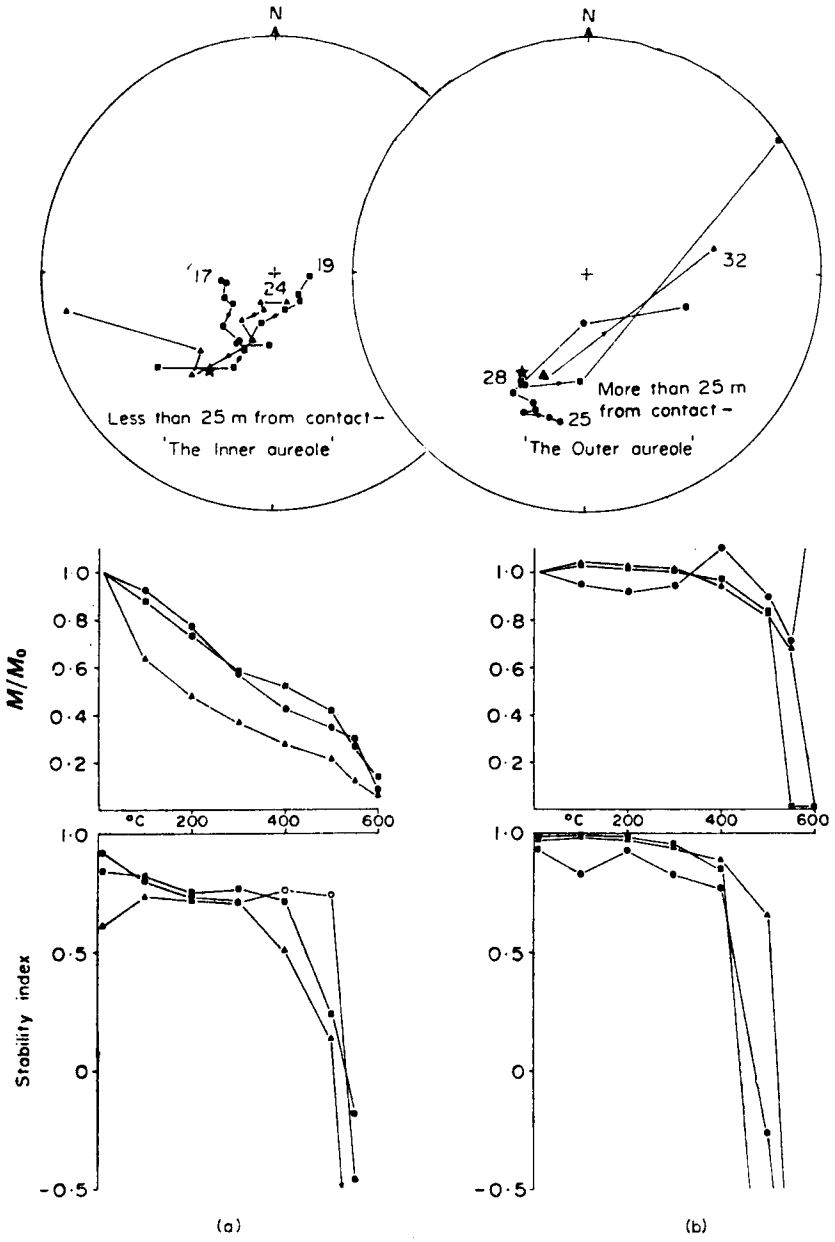


FIG. 3. Variation of direction (top), intensity (middle) and Stability Index (bottom) during thermal demagnetization of specimens from the Arrochar contact aureole, (a) less than, and (b) more than 25 m from the igneous contact. The mean stable RM of the complex (Briden 1970) is denoted by a star.

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remanent intensity and initial susceptibility. The Koenigsberger ratio (Q_n) varies erratically across the aureole (Fig. 2) so that remanent intensity is not proportional to initial susceptibility. This metamorphism is superimposed upon the Dalradian regional metamorphism and its mineralogical effect was principally to increase grainsize. If the iron oxides participated in this crystal growth, formerly small superparamagnetic grains may have been transposed into the ferromagnetic domain size region explaining the susceptibility variation.

The principal finding from thermal demagnetization, as described below, is that even the outermost samples, 170 m from the contact, appear to carry a total TRM in magnetite. Hence total intensity variation across the aureole must reflect these detailed changes in the magnetic mineralogy throughout the remanence-bearing phase, and not merely in the very low coercivity region which is reflected in measurements of χ.

The stability of natural remanence (NRM) against progressive thermal demagnetization (Figs 3 and 4) changes abruptly near 25 m from the contact, which is regarded as the divide between 'inner' and 'outer' parts of the aureole. This division is apparent in progressive AF demagnetization of typical specimens (Briden 1970), and in the generalized AF properties of the two zones given in Fig. 4. No such abrupt change is apparent in the total intensity and initial susceptibility profiles.

In both the inner and outer parts of the aureole the thermal and AF demagnetization curves can be matched (Fig. 4) and the remanence classified according to critical blocking temperature (T_B) and remanence coercivity (H_{CR}). The inner aureole shows T_B to be fairly uniformly distributed from present day surface temperatures to above 600 °C. About 20 per cent of the remanence has T_B less than 100 °C and H_{CR} less than 20 Oe. Another 40 per cent falls in the ranges 100 °C < T_B < 400 °C and 20 Oe < H_{CR} < 100 Oe; the next 30 per cent of remanence has T_B in the ranges 400–570 °C, and H_{CR} between 100 and 550 Oe. Finally some 10 per cent of the remanence has higher critical blocking temperatures than about 570 °C and remanence coercivities above 550 Oe, with an appreciable fraction exceeding 1000 Oe. In the outer aureole, the remanence spectrum is different. Intensity reduces

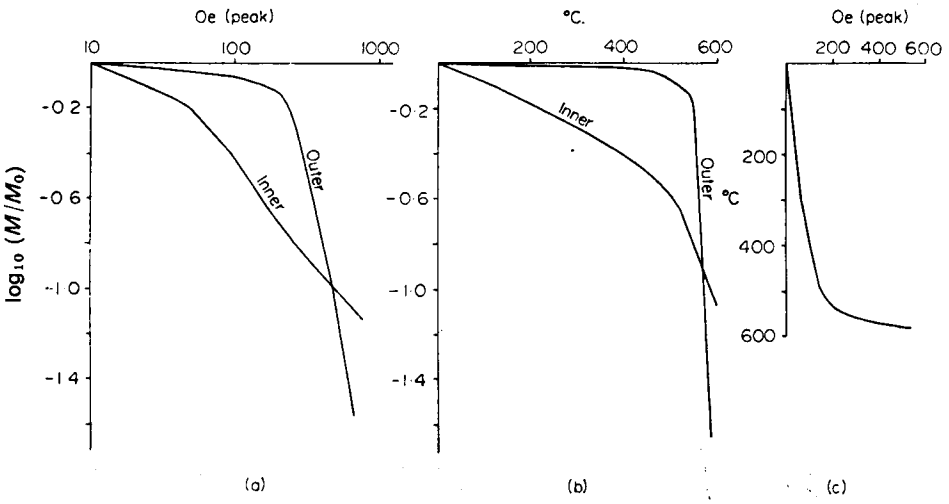


FIG. 4. (a) Mean AF demagnetization curve for the inner and outer aureole of the Arrochar Complex, replotted with logarithmic axes from Briden (1970, Fig. 4(b)); (b) mean thermal demagnetization curves for the inner and outer aureole replotted with logarithmic ordinate from Fig. 3; (c) approximate correlation of remanence coercivity and critical blocking temperature inferred by equating corresponding points in (a) and (b).

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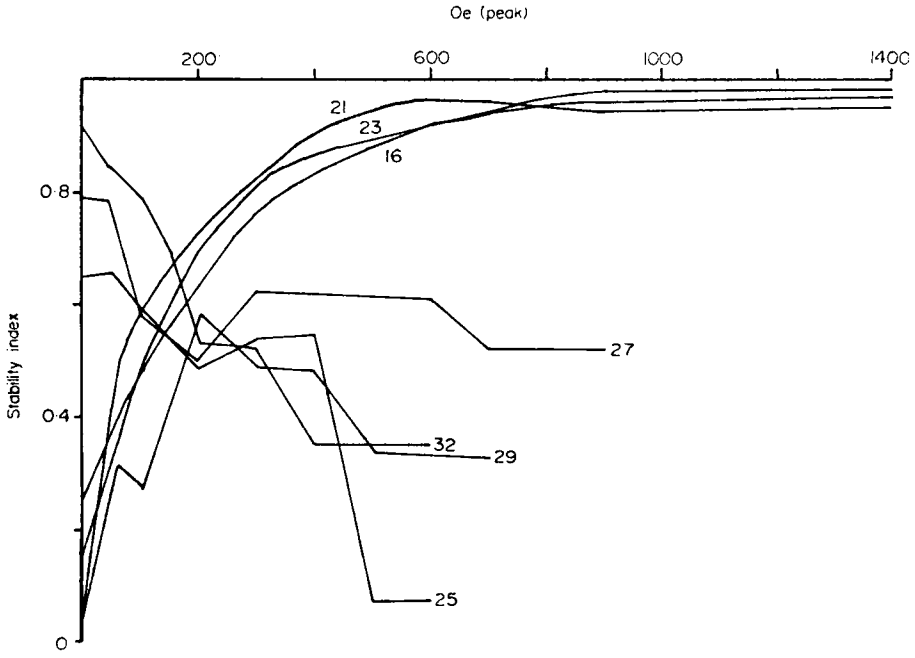


FIG. 5. Variation of SI during AF demagnetization. The values at $H = 400$ Oe have been used to produce the generalized profile at the bottom of Fig. 2.

by 10 per cent on demagnetization at 475 °C or 80 Oe, and falls to 10 per cent of total at 570 °C or 500 Oe. None of the systematically directed NRM survives heating to 600 °C, nor are coercivities greater than 700 Oe evident.

The Dalradian rocks in this area consist of a monotonous series of quartz-biotite-chlorite schists, so the magnetic variations across the aureole are probably due to the thermal metamorphism itself rather than any pre-existing systematic variation of lithology. A component with critical blocking temperature above 600 °C, and coercivities extending above 1000 Oe, is formed near the contact, which is not observed in the outer aureole. The main remanence carrier exhibits different spectra of coercivities (below 550 Oe) and blocking temperatures (below 570 °C) in the inner and outer zones. These spectra can be matched in the two zones to produce the correlation between critical blocking temperature and remanence coercivity illustrated in Fig. 4(c). Up to about 500 °C a linear relationship appears to hold, but at higher temperatures the coercivity dependence is more sensitive. The relation $T_B \alpha \log H_{CR}$ empirically describes the whole range of observations in Fig. 4(c). It is notable that the maximum T_B and H_{CR} correspond to the maximum observed values in magnetite or Ti-poor titanomagnetite. Petrological investigation confirms that this 'magnetite' is the commonest opaque phase throughout, and it is therefore likely to be the principal carrier of remanence. It occurs both in the form of euhedral, subhedral or skeletal grains, and as elongate needles aligned in the schistosity. Although the elongate magnetite could have high coercivities, this would not explain the high critical blocking temperatures exceeding the maximum Curie Temperature attained in the ulvospinel–magnetite series. This second remanence component must therefore be haematite or maghemite, but no such minerals have been observed in the inner aureole where their presence is also suggested by stability evidence. No maghemite has been observed in any of the rocks of the aureole and paradoxically haematite is only abundant in the *outer* part of the aureole where it does not appear to be an important remanence carrier. It occurs as large well-formed crystals, presumably

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too large to carry stable remanence. A similar situation was reported in the igneous rocks of the Arrochar *intrusion* (Briden 1970) where although haematitic alteration of titanomagnetite was occasionally observed it was not found in *all* the rocks which showed a high Curie point phase. Briden concluded that the high Curie point phase took the form of fine peripheral alteration of ilmenite and magnetite and the same may be true of the inner aureole.

The variation of Stability Index (SI, Briden 1972) across the aureole during thermal (Fig. 3) and AF demagnetization (Fig. 5) shows the distinction between the inner and outer aureoles in a different way. In thermal demagnetization, at temperatures up to 500 °C, SI is only moderate in the inner aureole, reflecting the distributed demagnetization curve, but is high in the outer aureole, in keeping with the more rectangular demagnetization curve. Measurements above 600 °C are inadequate to show the stable high temperature remanence in the inner aureole though its presence is evident (Fig. 3(a)). During AF demagnetization of the inner aureole, SI progressively increases as the main remanence carrier is removed at a progressively diminishing rate. It reaches near maximum values (0·9–1·0) above 600 Oe when the entire magnetite phase has been demagnetized but the stable high coercivity haematite has not been affected. In the outer aureole, the progressive removal of the entire remanence by AF treatment is reflected in moderate stability indices. No precise temperature calculations are possible without knowing the shape of the intrusion, but judging from the profiles calculated by Jaeger (1957) for temperatures adjacent to a plane sheet, 600 °C may be exceeded at a distance from a contact equal to perhaps 10 per cent of the thickness of an intrusion of intermediate composition and melting characteristics. Our farthest record is no more than 170 m from the contact of the intrusion and would imply a thickness of the order of 1·7 km which is rather less than the average diameter of the main Arrochar mass. These rough estimates are consistent with the palaeomagnetic observations which indicate that the magnetite Curie temperature was exceeded 170 m from the exposed contact.

The results described here show beyond doubt that the NRM in the contact aureole is associated with intrusion of the Complex. The mean remanence direction in the aureole agrees with those determined for the Arrochar igneous complex, and for the nearby Garabal Hill–Glen Fyne complex which shows precisely the opposite polarity (Briden 1970).

4. Mweelrea Ignimbrites

Deutsch (1969), Deutsch & Somayajulu (1970) and Murthy & Deutsch (1971) report palaeomagnetic results from the mid-Ordovician Mweelrea ignimbrites of the South Mayo Trough. They show that the ignimbrites have high magnetic stability and by positive conglomerate and fold tests, that the magnetization is pre-tilting, but the direction of this magnetization is anomalous when compared to British rocks of a similar age (see Papers I–III).

4.1 *Geological outline*

The Mweelrea ignimbrites outcrop in the area around Killary Harbour, Western Eire mapped by Stanton (1960), McKerrow & Campbell (1960) and Dewey (1963) (Fig. 6). Stanton (1960) describes the generalized internal structure of the six ignimbrite bands noting that they typically exhibit green tuffaecous bases, highly welded and columnar-jointed central portions, with less well-consolidated tops. He correlates the thickness of the green base with water depth at the time of ignimbrite formation and infers a slight north-westerly increase in water depth. This also supports the idea that depositional dips were very low and almost zero (G. H. Wilson 1972, private communication). Each ignimbrite sheet represents a sudden burst of igneous activity recording an instantaneous time plane over the depositional basin.

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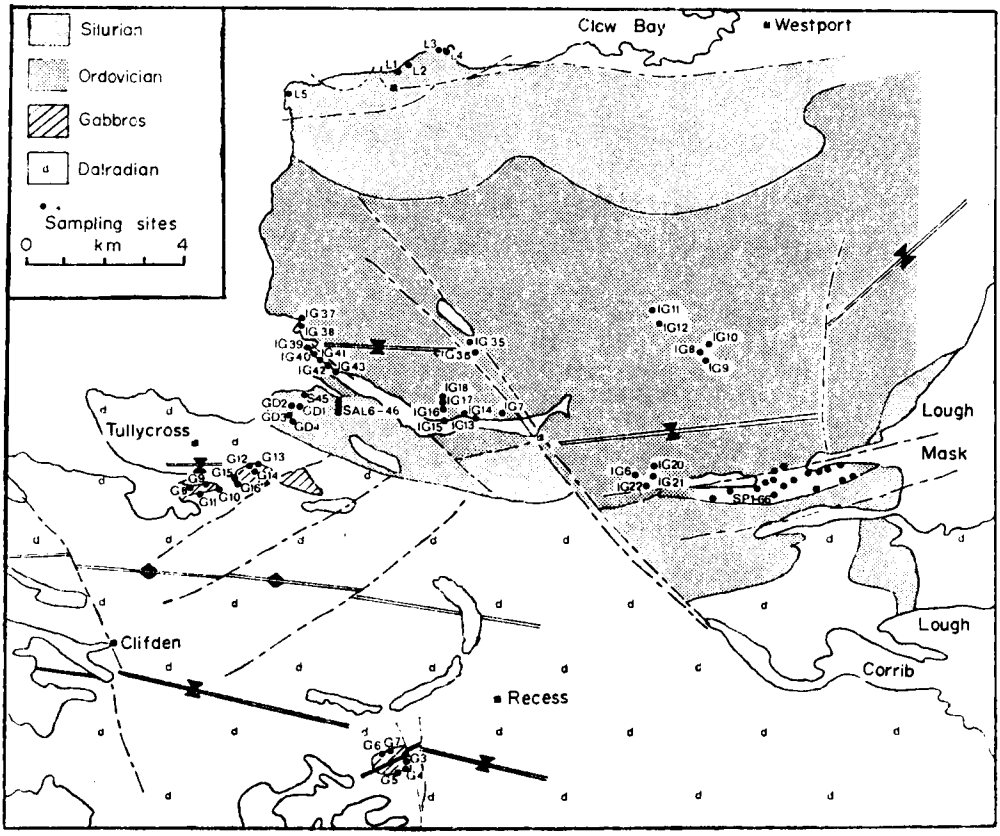


FIG. 6. Generalized geological map of Co. Mayo and Co. Galway, Eire, showing palaeomagnetic sampling localities. SP—spillites (Arenig), G—Connemara Gabbros, IG—Mweelrea Ignimbrites, GD—microgranodiorite, SAL—Salrock Formation, L—Knocknaveen Group.

The complex sedimentary environment is exemplified by the convergence in different parts of the region of bands 3 and 4, and 4 and 5. Dewey (1963) showed that the grit wedge between ignimbrite bands 5 and 6 varies from 1600 m on the northern limb of the Mweelrea syncline in Central Murrisk to 60 m on the southern limb in Joyces Country. McKerrow & Campbell (1960) working in Joyces Country point out that in nearly every case the ignimbrite sheets increase in thickness towards the south-east and suggest a source area in that direction. The most reliable evidence of age is given by Williams (1972) who suggests a mid-Llanvirn age for the ignimbrites on a brachiopod assembleage collected close to site 39.

4.2 *Total NRM and AF demagnetization*

Within-site precision of total NRM directions is here estimated by Fisher (1953) analysis, giving unit weight to each specimen (Table 1). By this criterion only four of our 25 site directions are random at the 95 per cent confidence level (Watson 1956). The directions at 17 sites form a major group around $D = 139°$, $I = 42°$; four sites, all from band 3 form a minor group near $D = 330°$, $I = 45°$. There is no systematic intensity variation between bands, and the intensity range is large—from $6{\cdot}7 \times 10^{-6}$ Gauss to $3{\cdot}5 \times 10^{-3}$ Gauss.

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Table 1

Mweelrea ignimbrites-site data

Ignimbrite band	Site	Total NRM N (specs)	R	k	in situ D	in situ I	Dip corrected D	Dip corrected I	Optimum field	AF cleaned N (specs)	R	k	in situ D	in situ I	Dip corrected D	Dip corrected I
1	7	4	3·99	623·0	131	−27	132	+23	200	3	2·99	677·7	131	−33	135	+17
	11	6	5·64	13·9	127	+48	136	−13	500	6	5·29	7·0	112	+42	123	−13
	13	10	8·13	4·8	146	+ 9	131	+61	300	6	5·65	14·1	174	−4	181	+49
	35	6	4·59	3·6	156	+21	156	−29	400	7	5·73	4·7	132	+25	133	−21
	37	6	5·94	80·0	350	−17	350	−5	200	6	5·33	7·5	41	−10	40	−4
	38	6	5·27	6·8	61	+47	74	+48	300	7	6·82	33·5	75	+27	81	+26
	43	6	3·32	not significant					200?	5	3·50	not significant				
2	14	9	5·19	2·1	129	+ 1	113	+44	300	7	5·81	7·6	146	−18	148	+27
	36	6	5·40	8·4	305	+75	166	+53	500	7	6·66	17·6	348	+71	145	+59
3	15	4	3·70	10·1	173	+80	337	+45	600	4	3·83	17·7	186	+83	336	+41
	16	9	7·90	7·3	337	+74	339	+35	800	6	5·44	8·9	166	+87	350	+49
	20	7	5·38	3·7	181	−14	199	+47	800	5	4·68	7·5	227	+63	306	+42
	40	6	5·06	5·3	304	+23	311	0	200	4	3·16	3·6	306	+ 6	303	−15
	41	5	3·07	not significant					200	4	3·50	6·0	355	−10	352	−47
	42	6	1·68	not significant					300	4	3·28	4·2	262	+12	269	+17
4	6	3	2·99	1616·0	141	−12	136	+40	700	3	2·99	1125·6	141	−17	138	+35
	8	10	9·95	194·0	117	+52	138	+ 7	200	7	6·98	237·7	115	+51	136	+ 6
	10	10	9·67	27·1	120	+70	141	+36	200	7	6·98	241·1	128	+57	144	+22
	12	9	8·97	294·8	34	+68	133	+45	200	6	5·99	603·1	62	+70	133	+34
	17	9	5·68	2·4	120	−47	133	−10	200	7	3·48	not significant				
	21	12	11·03	11·3	129	−15	126	+39	200	8	7·70	23·1	141	−22	140	+33
	39	6	4·32	3·0	105	+ 7	121	+44	200	6	4·38	3·1	136	−11	135	+15
5	9	10	8·87	8·0	319	+84	160	+36	300	7	6·71	20·3	65	+63	126	+27
	18	5	4·07	4·3	170	+17	160	+55	200	7	5·54	4·1	170	+15	167	+31
	22	6	1·45	not significant					200	6	5·51	10·2	125	+12	105	+63

Table 2

Mweelrea Ignimbrites—Fold tests and overall analysis

Band	N	*in situ* R	*in situ* k	Dip corrected R	Dip corrected k	D	I	Virtual geomagnetic pole Long	Lat	$\partial\psi$	$\partial\chi$	Significance of fold test
1	6 sites	3·96	2·5	4·11	2·6	114	+12	58° E	9° S	27	52	Not significant
	35 specs	21·31	2·5	21·4	2·5	110	+12	60 E	7 S	10	20	Low precision
2	2 sites	Insufficient for analysis										
	13 specs	9·15	1·5	11·57	8·4	147	+45	20 E	5 S	12	19	Significant at 99 per cent level
3	6 sites	3·67	2·1	4·18	2·8	318	+18	315 E	19 N	27	52	
	32 specs	17·45	2·1	19·75	2·5	315	+19	226 E	33 N	11	21	Not significant low precision
4	6 sites	4·45	3·2	5·89	45·0	137	+24	33 E	14 S	6	11	
	37 specs	26·05	3·3	34·30	14·0	138	+24	33 E	15 S	4	7	Significant at 99 per cent level
5	3 sites	Insufficient for analysis										
	20 specs	12·01	2·6	16·28	5·1	136	+43	29 E	3 S	12	20	Significant at 95 per cent level
All sites	23 sites	10·39	1·8	11·49	1·9	128	+41	36 E	1 S	23	38	
	5 bands	3·25	Non-significant									
Bands	17 sites	11·29	2·8	14·17	5·6	131	+27	38 E	11 S	10	18	Significant at 95 per cent level
1, 2, 4, 5	4 bands	3·83	17·9	3·84	18·6	133	+30	36 E	10 S	13	24	Not strictly applicable

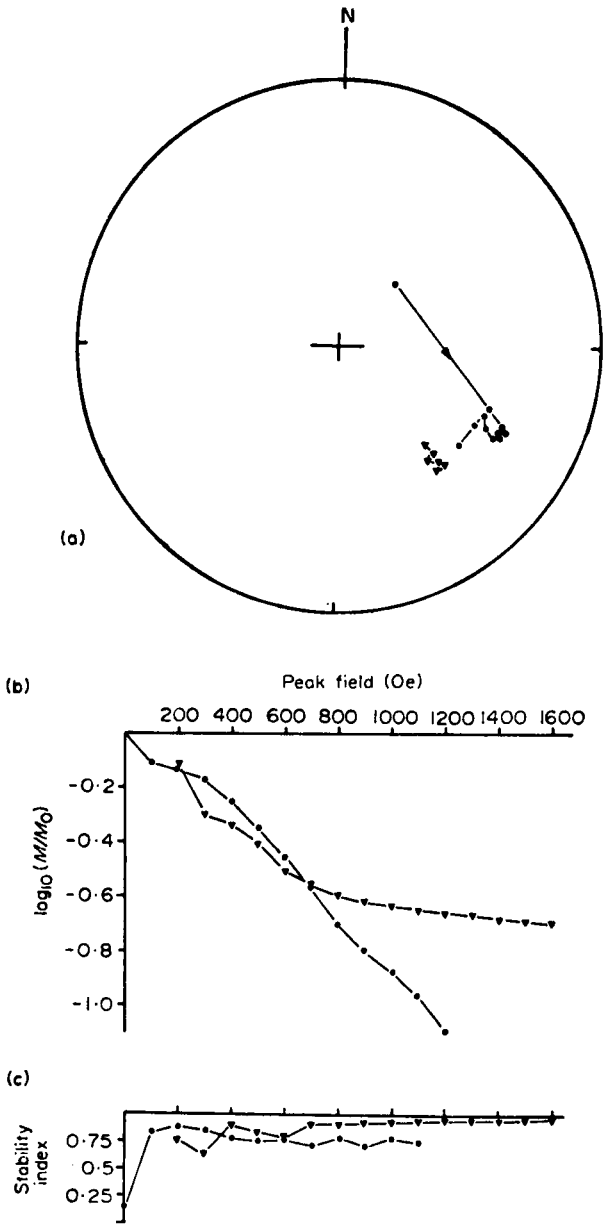


FIG. 7. Mweelrea ignimbrites: A.F. demagnetization characteristics showing (a) direction change, (b) intensity change and (c) Stability Index.

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The progressive AF demagnetization characteristics agree with the results of Deutsch & Somayajulu (1970). Cleaning merely removes low-coercivity components, oriented close to the present geomagnetic field (Fig. 7). High stability of direction and intensity suggests the presence of haematite (N. D. Watkins, in Deutsch & Somayajulu (1970)), in addition to the titanomagnetite which is the principal remanence carrier.

4.3 *Magnetic cleaning*

Of the 25 sites, 13 were cleaned at 200 Oe, five at 300 Oe, and the rest in slightly higher fields. Because it was not possible to select an optimum field for site 43, it is discounted from further analysis. After cleaning 23 site mean directions are significant, sites 17 and 43 remain random. Sites 6, 7, 8 and 12 which were well defined prior to cleaning show little or no change of direction and precision after cleaning; the precision at sites 10 and 38 increases by eightfold and fivefold respectively during cleaning.

A fold test applied to site mean directions from bands 1, 2, 4 and 5 is significant at the 95 per cent level (but when this test is performed on *band* mean directions it appears to be non-significant because the effect of folding is largely obscured by the band-mean calculation). The presence of pre-deformation tilting caused by basin-sedimentation would be eliminated by tilt correction only if the basin and the deformation had the same horizontal axial trend. Evidence from the interbedded sediments shows that the basin deepened to the north and west during the period of ignimbrite eruption. The procedure of applying a single tilt-correction, ignoring these early gentle warpings, introduces errors of less than 10° in the final direction estimation at individual sites, and these errors are not likely to be highly systematic. Therefore this simplified approach is adequate. Fold tests can be applied to each band individually and produce positive results for bands 2, 4 and 5. Poor precision and small dip variation of bands precludes the application of decisive fold tests for bands 1 and 3 (Table 2). Because band 3 is bounded above and below by bands carrying primary remanence, a secondary magnetization is unlikely to be restricted to band 3 alone. Therefore it is argued that the remanence found in band 3 is primary (or at least older than band 4 *and* the major folding) and hence that band 3 records an instantaneous anomalous local geomagnetic field.

It is not possible to define the time interval between successive ignimbrites. The directions from bands 1, 2, 4 and 5 all fall within the south-east quadrant, the inclination is downward varying from shallow to moderately steep (Table 2). Band 3, although fulfilling all the same stability criteria as the other bands and not showing any marked difference in intensity, gives a downward direction in the north-west quadrant. The difference between bands 3 and 4 disagrees with the result of Deutsch & Somayajulu (1970). It would seem that their statistical test actually applied to two units of band 4, which our survey independently shows give very similar directions. After AF cleaning at the chosen field, the range of intensities was markedly reduced—$1{\cdot}6 \times 10^{-6}$ Gauss to $0{\cdot}9 \times 10^{-6}$ Gauss—but there is still no systematic variation between bands. At site 35, where samples were taken systematically through band 1, sequential changes in intensity and direction are not observed in the profile.

4.4 *Discussion*

Apart from their erroneous identification of Band 3, the results of Deutsch & Somayajulu (1970) are identical with ours in all major aspects. The directions estimated for each band agree (notably band 4 where we have $D = 137°$, $I = 24°$, $k = 45{\cdot}0$ and Deutsch & Somayajulu have $D = 135°$, $I = 22°$, $k = 45{\cdot}0$).

Since band 3 is anomalous it is not included in the overall analysis. Giving unit

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weight to each site the final result of $D = 131°$, $I = 27°$, $k = 6$, is statistically the same as that quoted by Deutsch & Somayajulu (1970), namely $D = 135°$, $I = 36°$, $k = 12$. Because of the unequal distribution of sites over the five bands sampled, it may be more correct to assign unit weight to each band. This analysis gives a final result $D = 133°, I = 30°, k = 19$ which corresponds to a pole at Long 36°, E Lat 10° S; Deutsch & Somayajulu (1970) calculated their final result as $D = 134°$, $I = 32°$, $k = 23$, in the belief that they had sampled four bands. Recalculating on the basis of three bands which they had actually sampled gives $D = 135°$, $I = 37°$, $k = 25$ so the basic agreement between the two studies remains. Hence the anomalous palaeomagnetic direction in the Mweelrea ignimbrites is completely verified. It remains to determine the duration and areal extent of this anomaly. There are many rock types exposed in the South Mayo Trough and adjacent regions suitable for investigating this problem. Studies of these other rocks are not yet complete but some provisional results are reported in Sections 5–9.

5. Arenig Pillow lavas, Lough Nafooey

The basal Ordovician of the South Mayo Trough (Fig. 6) is represented by a series of pillow lavas, with interbedded tuffs and felsite breccias. Gardiner & Reynolds (1912, 1914), mapping the area in great detail, dated these rocks as Arenig on the basis of fossiliferous shales occurring as pockets within the lavas. As further west (Section 4) the structural history of this area is controlled by major E–W trending faults. The pillow lavas were folded into a major anticline with this same E–W axial trend. The detailed structure may be more complex than was originally thought, since cherts interbedded with lavas on Knock Kilbride exhibit minor folds with near vertical axes yet an adjacent block has only a gentle northerly dip. Twenty-two sites have been collected from the pillow lavas on both limbs of the anticline by field drilling and hand sampling. In this section we report the total NRM results of this collection. Five sites are non-significant at the 95 per cent level and so are excluded from further consideration. The remaining 17 combine to give a mean of $D = 139$, $I = +83$ ($k = 6{\cdot}6$, $\alpha_{95} = 15°$ *in situ*), and $D = 355°$, $I = 58°$, $k = 2{\cdot}6$ (dip corrected). Hence a fold test of remanence is negative. Progressive AF demagnetization shows that all sites, even those with high NRM precision, show large direction and intensity changes; and a conclusive palaeomagnetic result must await full AF treatment.

6. Connemara Gabbros

The Upper Cambrian/Lower Ordovician gabbros were intruded into regionally metamorphosed Dalradian basic lavas and sediments (Fig. 6). Metamorphic aureoles were produced in the adjacent rocks, but they were themselves metamorphosed and deformed at a later date by continued regional metamorphism and compression. Leake (1970) suggests the gabbros were intruded syntectonically with the F_3 deformation and affected by the subsequent F_4 deformation. They have been dated by Rb/Sr analysis of separated zircons at 510 ± 10 My (Pidgeon 1969).

A provisional result is reported from 14 sites collected from three bodies; Dawros—4 sites, Currywongaun—5 sites and Cashel—5 sites. Two of the total NRM results are not statistically significant at the 95 per cent level (Watson 1956). When dip-corrected the majority of sites fall into a group round $D = 225°$, $I = 20°$, sites 3, 4, 5 and 16 do not fall in this group but give *in situ* directions very close to the present geomagnetic field. Although the majority of specimens are strongly magnetized the intensity range is wide, extending from 23×10^{-3} Gauss to $0{\cdot}05 \times 10^{-6}$ Gauss.

Pilot specimens from each site were treated by progressive AF demagnetization (Fig. 8). Upon demagnetization some specimens (from sites 4, 8, 10, 13, 14 and 16) show little (or no) direction change and regular intensity decay. Others (e.g. site 6)

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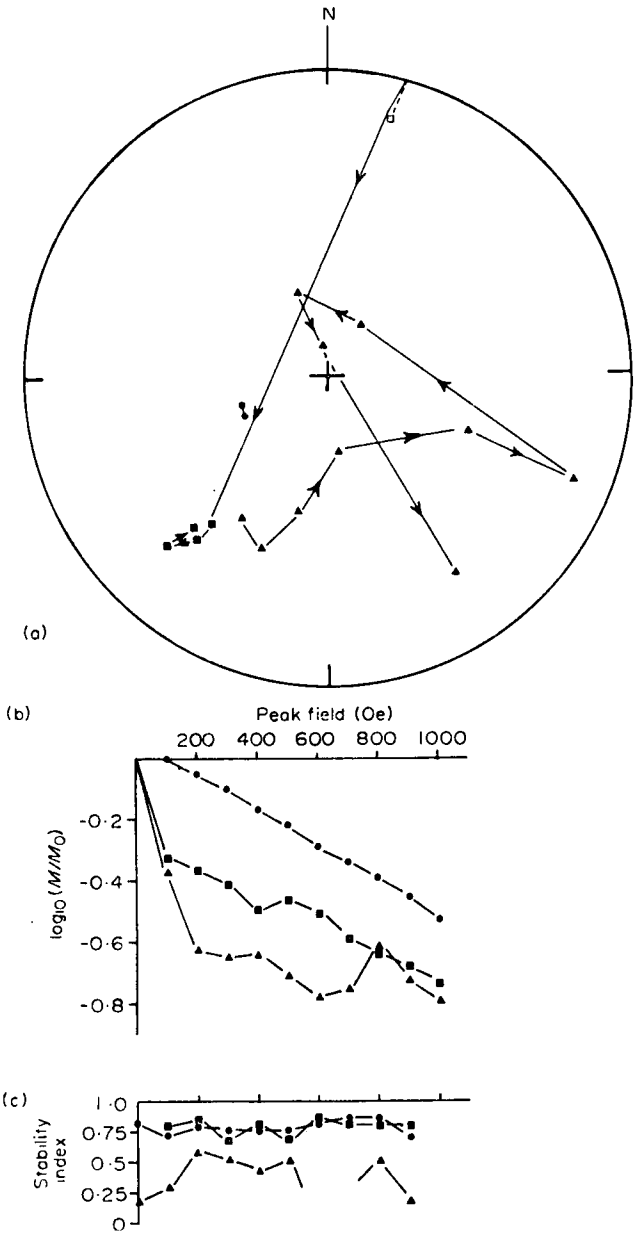


FIG. 8. Connemara gabbros: progressive AF demagnetization characteristics (a) direction changes, (b) intensity change, and (c) Stability Index.

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Table 3

Connemara Gabbros—site data

		Total NRM							A.F. cleaned						
	Site	*N* (specs)	*R*	*k*	*D* (*in situ*)	*I* (*in situ*)	*D* (dip corrected)	*I* (dip corrected)	Peak field (Oe)	*N* (specs)	*R*	*k*	*D* (*in situ*)	*I* (*in situ*)	*D* (dip corrected)	*I* (dip corrected)
CASHEL HILL	3	5	4·33	5·9	348	+70	355	+15	200	5	4·45	7·2	44	+67	13	−1
	4	6	5·73	18·9	51	+88	4	+9	300	6	5·67	15·3	225	+62	340	+30
	5	6	5·76	21·1	116	+83	164	+48	200	6	5·66	14·7	302	+81	186	+57
	6	4	3·96	75·3	233	+37	215	+23	200	5	4·77	17·3	231	+31	217	+17
	7	6	5·62	13·1	238	+39	220	+22	500	5	3·88	3·5	257	+28	228	+37
DAWROS	8	5	4·56	9·0	217	−27	213	+38	200	5	4·72	14·3	221	−14	205	+50
	9	6	5·62	13·1	41	+30	225	+12	500	5	4·73	14·9	226	−3	224	+26
	10	5	3·97	3·8	241	−8	239	+25	200	4	3·97	101·2	225	−8	223	+20
	11	6	2·34		not significant				200	5	4·92	51·1	226	−15	219	+18
CURRYWONGAUN	12	5	3·49		not significant				500	4	3·66	8·9	215	0	260	+46
	13	6	5·88	42·5	197	−4	250	+66	100	6	5·91	54·5	195	−10	209	+59
	14	6	5·97	158·2	208	−13	237	+50	100	6	5·95	100·0	212	−13	228	+48
	15	6	5·68	15·8	88	+44	67	+11	400	6	5·38	8·1	201	+43	180	+63
	16	6	5·85	34·3	66	+67	46	+24	300	6	5·76	21·1	173	+77	82	+69

Table 4

Connemara Gabbros—Analysis of site mean data—AF cleaned

								Palaeomagnetic Pole		
		N	*R*	*k*	*D*	*I*	Long	Lat	$\partial\psi$	$\partial\chi$
CASHEL HILL	*in situ*	5 sites	4·46	7·4	246	+65	307° E	24° N	39	49
		27 specs	21·90	5·1	244	+66	309 E	25 N	18	22
	dip corrected	5 sites	2·55		not significant					
		27 specs	12·68	1·8	272	+62	288 E	34 N	36	47
DAWROS	*in situ*	4 sites	3·99	193·9	225	−10	296 E	30 S	3	7
		20 specs	19·10	20·5	225	−10	297 E	31 S	4	8
	dip corrected	4 sites	3·88	24·8	219	+29	312 E	14 S	11	21
		20 specs	17·80	14·8	219	+29	312 E	14 S	6	10
CURRYWONGAUN	*in situ*	5 sites	4·04	4·2	204	+18	324 E	24 S	23	44
		27 specs	20·46	4·0	203	+21	326 E	23 S	9	17
	dip corrected	5 sites	4·51	8·2	217	+67	326 E	17 N	39	47
		27 specs	24·17	7·0	216	+67	330 E	17 N	15	18
All Bodies	*in situ*	14 sites	10·22	3·4	221	+25	309 E	15 S	15	27
		72 specs	49·70	3·2	219	+26	311 E	15 S	7	12
	dip corrected	14 sites	10·23	3·4	227	+53	312 E	7 N	24	35
		72 specs	50·88	3·2	226	+56	314 E	8 N	12	16

behave similarly once a large low coercivity component has been eliminated. A third class exhibit more complex demagnetization curves, the intensity decaying steeply and irregularly, associated with large and random direction changes. All sites were cleaned in peak fields less than 600 Oe of which 10 sites were treated in 300 Oe or less. After AF treatment all 14 sites are significant at the 95 per cent level (Table 3); sites 11 and 12 which had non-significantly grouped NRMs become significant after cleaning.

The magma probably did not cool below its Curie point until after the completion of the F_3 deformation. The tilt correction for igneous lamination applied in Table 3 by rotating about its present strike accounts principally for the effects of F_3 folding. Residual effects of F_4 have not been corrected for. Table 4 shows the AF cleaned mean results for each gabbro mass and the overall combined result.

The *in situ* directions from the Cashel and Dawros masses are more closely grouped than the dip corrected. Tilt correction of the Currywongaun result produces a small numerical increase in precision. Whether or not tilt correction is applied the direction is near 220° with positive inclination; hence there are two alternative interpretations of the present results. First, if the direction recorded by the gabbros is primary remanence of early Ordovician age, then this would suggest a 40° *clockwise* rotation of the gabbro masses by comparison with results from Great Britain (Papers I–III). The Mweelrea ignimbrites in the adjacent Lower Palaeozoic sedimentary basin record a 40° *anticlockwise* rotation, implying that both these rotations are of local extent. On this interpretation the precise time relation between F_3 deformation and magnetization remains uncertain. The tilt-corrected inclination is 53° and the *in situ* inclination 25°; these are to be compared with inclinations in other Ordovician rocks in the British Isles which range from 27° to 55°. Second, either the dip-corrected or *in situ* result might be considered a true record of primary unrotated remanence.

7. Salrock Formation sediments and associated intrusives

7.1 *Geology and sampling*

The youngest Lower Palaeozoic rocks found in the South Mayo Trough are a 1000 m thick sequence of sandstones, siltstones and shales (the Salrock Formation), outcropping on the south side of Killary Harbour (Fig. 6), and considered by McKerrow & Campbell (1960) to be of Wenlock age, since they conformably overlie middle Wenlock sediments (the Upper Owenduff Group). Intruded into the Upper Owenduff Group are two microgranodiorite sills and intruded into the Salrock Formation are three sills of andesite. The Upper Owenduff Group and the Salrock Formation were tilted steeply to the north and overthrust by the Leenane Grits (Arenig) near the end of the Silurian.

7.2 *Salrock Formation sediments—NRM and site mean analysis*

Of 32 total NRM site means in the Salrock Formation sediments four are statistically random at the 95 per cent level (Table 5). The declinations (dip-corrected) of all but two sites are within the north-east quadrant, but inclinations vary from $+37°$ to $-37°$ (Fig. 9). Attempts to clean the red beds using AF demagnetization produced only small changes of direction and intensity, and suggest that haematite is the remanence carrier. A test of stability is based on the observation of similar directions from both the red and green sediments. The green rocks of site 15 show no obvious haematite staining or secondary alteration, and the AF demagnetization characteristics eliminate members of the ilmenite–haematite series as possible remanence carriers. There is a rapid decay of intensity and only 10 per cent of the original remanence is left after cleaning at 400 Oe. This demagnetization produces a declination change, but the cleaned direction remains within the group of directions observed in the red beds.

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Table 5

Salrock Group Sediments—Site mean analysis: Total NRM

	N(sites)	R	k	D	I	Palaeomagnetic Pole Long.	Lat.	$\partial\psi$	$\partial\chi$
in situ	28	24·2	7·2	71	+49	75° E	35° N	10	14
dip corrected	28	24·5	7·8	49	+1	115 E	23 N	5	10

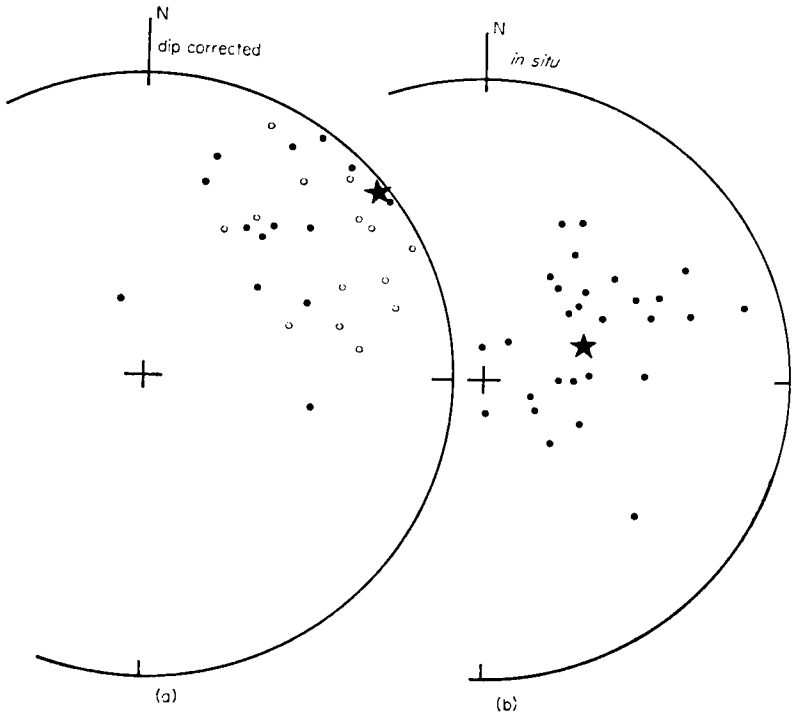


FIG. 9. Salrock group sediments. Total NRM site mean directions (a) dip corrected, (b) *in situ*.

7.3 *Intrusives: NRM and site mean analysis*

Total NRM site means in the microgranodiorite are poorly defined (two sites are random and two have $k < 4{\cdot}5$). Of the three sites (SAL 4, 43 and 45) from the andesite sills, two (SAL 43 and 45) have high within-site precision. The five significant sites from the intrusives are grouped round $D = 340°$, $I = +30°$ after dip correction. The andesites and the microgranodiorites (Fig. 10) show a regular decay of intensity, little direction change and hence high and uniform Stability Indices. The microgranodiorites were cleaned at 300 Oe (peak) and the andesites were cleaned between 200 and 600 Oe each. After cleaning all seven sites are significant (Table 6). The microgranodiorite directions have been combined with the andesite directions to obtain an overall mean in Table 6 on the assumption that the two groups of intrusives are of similar age.

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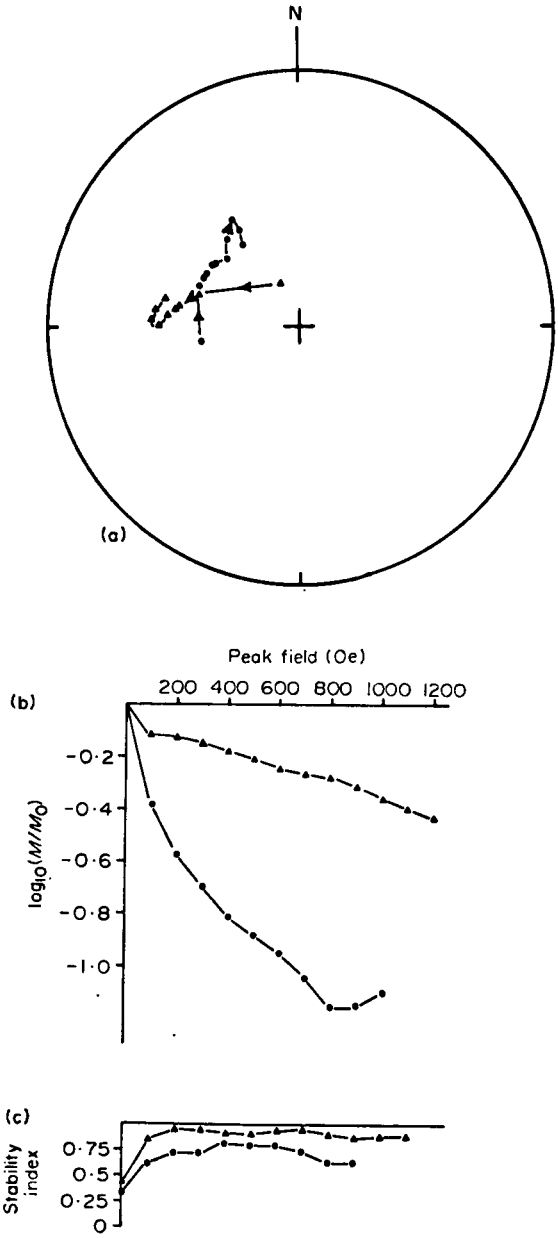


FIG. 10. Intrusives in Salrock Group: microgranodiorite (circles), andesite (triangles), Progressive AF demagnetization characteristics (a) direction, (b) intensity, and (c) Stability Index.

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Table 6

Intrusives and baked contacts of Upper Silurian age, Killary Harbour: site data

Site	Total NRM							AF cleaned								
	N (specs)	*R*	*k*	*D* *in situ*	*I* *in situ*	*D* dip corrected	*I* dip corrected	Peak field	*N* (specs)	*R*	*k*	*D* *in situ*	*I* *in situ*	*D* dip corrected	*I* dip corrected	Long Palaeomagnetic Pole	Lat Palaeomagnetic Pole
GD1	7	2·74	not significant					300	6	5·59	12·1	261	+57	316	+48	240° E	50° N
GD2	6	3·77	not significant					300	7	6·72	21·1	256	+30	283	+37	262 E	24 N
GD3	6	3·90	2·4	330	+75	3	+28	300	5	4·91	46·7	274	+17	291	+19	248 E	21 N
GD4	6	4·85	4·4	253	+62	320	+54	300	7	6·80	29·7	228	+48	317	+66	266 E	63 N
SAL4	9	6·92	3·8	258	+24	317	+49	200	7	5·32	3·6	262	+3	290	+40	309 E	47 N
contact	4	3·38	4·8	241	+37	348	+58	400	3	1·85	not significant						
								combined	10	6·90	2·9	252	+5	288	+49	265 E	34 N
SAL43	6	5·88	42·5	15	+78	26	+19	600	7	5·32	3·6	249	+43	332	+59	235 E	66 N
contact	6	5·81	26·1	247	+18	282	+55	100	7	6·88	48·3	244	+11	272	+52	280 E	27 N
								combined	14	11·69	5·6	246	+25	295	+58	269 E	44 N
SAL45	6	5·93	67·0	245	+69	344	+43	500	8	7·02	7·2	227	+4	244	+50	298 E	11 N
contact	6	4·07	2·6	207	+21	285	+68	100	7	6·50	12·1	213	+18	249	+63	303 E	24 N
								combined	15	13·43	8·9	220	+11	246	+56	300 E	17 N

Site mean analysis: AF cleaned

	N (sites)	*R*	*k*	*D*	*I*	Long	Lat (palaeomagnetic pole)	$\partial\psi$	$\partial\chi$
in situ	7	6·42	10·5	248	+29	286 E	0 N	12	22
dip corrected	7	6·60	15·0	291	+50	264 E	36 N	14	21

At each of the andesite sites samples were also collected from adjacent baked sediments. Applying the test of Watson (1956), it can be shown that the baked sediments do not give a significantly different mean direction from that recorded by the adjacent sill (Table 6). An exception to this is that the NRM in contact samples of site 4 is widely scattered. In the site mean analysis the contact and igneous samples are treated together at each site. A fold test of remanence is not significant ($P = 0{\cdot}05$) but tilt-correction does improve precision from 10·5 to 15·0.

7.4 *Discussion*

The remanence in the intrusions is likely to be primary because baked sediments have similar NRM directions, though unbaked sediments do not. The application of tilt-correction to the AF cleaned remanence in the intrusives and their contacts produces an appreciable increase in precision. They provide an estimate of the late Silurian geomagnetic field $D = 291°$, $I = +50°$, corresponding to a palaeomagnetic pole at 36° N, 264° E ($A_{95} = 17°$). This is anomalous in declination compared with other Siluro–Devonian data (Paper IV). (It might be noted that the *in situ* direction in these rocks is more similar to typical Devonian magnetizations and might be taken to imply intrusion after tilting.)

Total NRM in the Salrock Group sediments is grouped around $D = 71°$, $I = +49°$ *in situ*, and $D = 49°$, $I = +1°$ after tilt-correction. Neither of these mean directions is easily interpretable in terms of Silurian or younger geomagnetic fields, but the difference in magnetization between unbaked and baked parts of the sequences is an indication that at least part of the remanence is syngenetic or diagenetic.

8. Knocknaveen Group, Louisburgh

The northern margin of the South Mayo Trough is marked by a major fault system compared to the Highland Boundary Fault by Dewey (1969). North of the fault metamorphosed Dalradian sediments and volcanics crop out, and to the south are folded metamorphosed Silurian sediments; the fault line is marked by serpentinite bodies. Thrust over this contact from the north is a series of unmetamorphosed Wenlock red beds, the Knocknaveen Group (Phillips, Rickards & Dewey 1970). It has been suggested that these sediments, ranging from arkoses to fine siltstones, are the lateral equivalents of the Salrock Group of Killary Harbour. Folding is complex and fold tests of remanence are possible not only between sites spreading over some 15 km, but also within-site.

The five sites in this Group are probably not from the same thrust slice (Fig. 6). Site 2 was collected from a plunging minor fold where it is possible to correct for bedding-tilt in two stages. Total NRM results at the five sites are statistically significant at the 95 per cent probability level. Within-site grouping is good, but between-site grouping is poor, site 1 having steep positive inclination while sites 2, 3, 4 and 5 have negative inclinations. Progressive AF demagnetization was attempted on some specimens and, as is common in redbeds, treatment up to 1800 Oe (peak) produced little change in intensity or direction. The only exceptions to this were two samples from site 1 with NRM site mean direction diverging from the significantly grouped sites 2–5. With AF cleaning the direction of these site 1 samples moves into closer agreement with the directions recorded at sites 2–5.

The *in situ* analysis of four sites at $D = 197°$, $I = -68°$, $k = 4{\cdot}0$ does not match with any British Palaeozoic palaeomagnetic direction yet proposed. The tilt corrected result at $D = 34°$, $I = -54°$, $k = 9{\cdot}5$ (palaeomagnetic pole Long. = 143°, Lat. = 3° S, $d\psi = 31°$, $d\chi = 44°$) is supported by some AF demagnetizations and the fold test, and compares closely with Siluro–Devonian results from Great Britain.

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9. Ordovician igneous rocks in Pembrokeshire

9.1 *Trefgarn Andesitic Series*

Andesitic rocks of Arenig age at Treffgarne (4° 59′ W, 51° 53′ N (Fig. 11)) described by Thomas & Cox (1924) have been sampled by Almond (1971) in quarries where they dip 20°–30° NNE. Total NRM intensities range between 1 and 200×10^{-6} Gauss. Of 10 sites investigated seven show significantly grouped remanence before and after AF cleaning, at 600 Oe (Fig. 12(a)). Coercivity (Fig. 12(b)) increases with the oxidation state of the rocks and samples in which pyrite and subsidiary titanomagnetite are the principal opaque phases are less stable. Haematite-rich horizons retain 85–90 per cent of total NRM after demagnetization to fields of 1000 Oe or more. The lowest part of the sequence consists of primary lavas (sites 1 and 2), the remainder is locally reworked volcanoclastics of identical composition and presumably formed by erosion of these same lavas (Evans 1966). These sediments show current bedding and ripple marks and can be seen to be banked up against cliffs of the lavas. The *in situ* remanence is probably a better record of an ancient geomagnetic field direction for two reasons. First, much of the observed dip is depositional. Second, there is no distinction in intensity or direction of remanence between the rock types, and the remanence of the whole sequence is probably of diagnetic or later CRM origin. The most likely process to have operated is soda metasomatism contemporaneous with the Arenig igneous activity, as has been inferred in the Builth Volcanic Series (Paper I).

9.2 *Fishguard Volcanic Series*

Extrusives of Lower Ordovician age on the Pen Caer peninsula west of Fishguard have been sampled at 20 sites by Radford (1971) (Fig. 1). The study was confined to the middle (Spilite) division of the Fishguard Volcanic Series (Thomas &

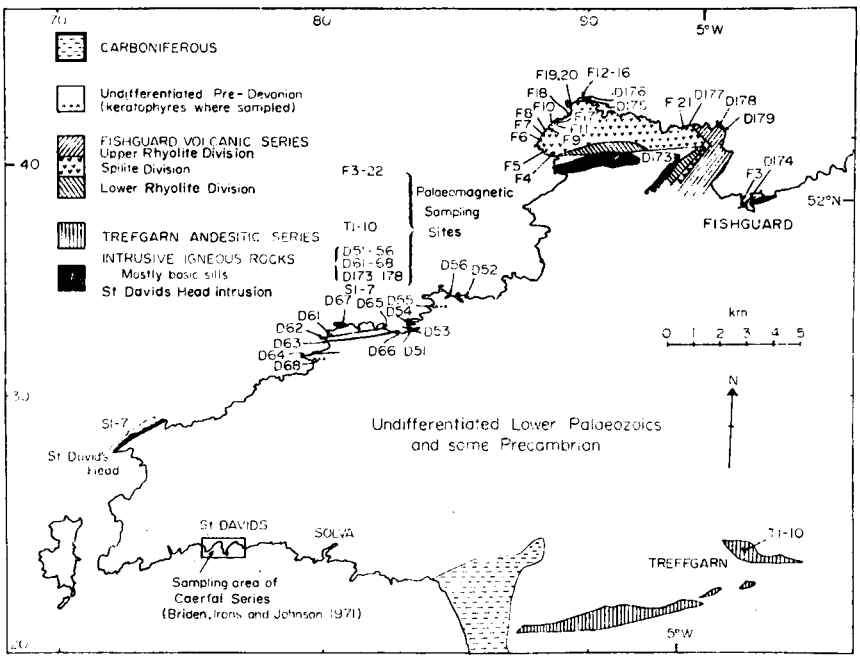


FIG. 11. Generalized geological map of northern Pembrokeshire showing palaeomagnetic sampling areas.

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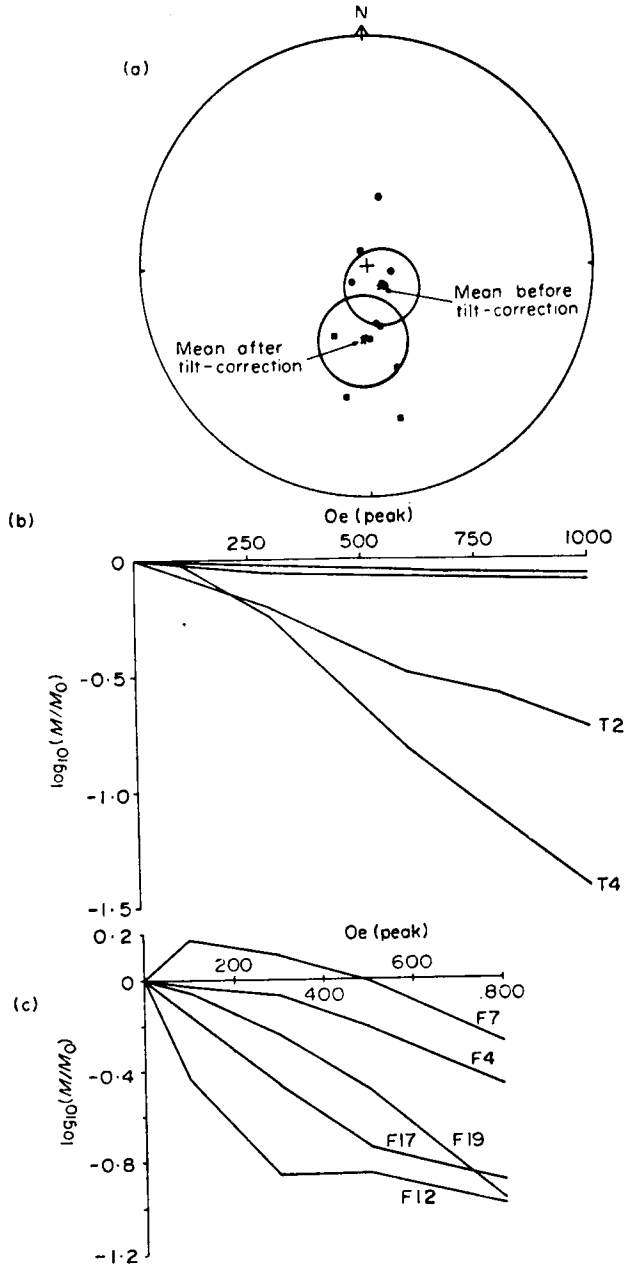


FIG. 12. Trefgarn Andesitic Series. (a) Site mean directions of RM after AF cleaning at 600 Oe, before (circles) and after tilt correction (squares), together with the overall mean and circle of 95 per cent confidence. (b) Typical AF demagnetization curves. (c) Fishguard Volcanic Series AF demagnetization.

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Table 7

Remanence directions in the Fishguard Volcanic Series (Spilite Division) after AF cleaning in a peak field 600 Oe

Site	Rock type	*N* (cores)	*R*	*k*	*D* (*in situ*)	*I* (*in situ*)	*D* (dip corrected)	*I* (dip corrected)	VGP Long (dip corrected)	VGP Lat (dip corrected)
3	Columnar flow	3	2·82	11·0	(115	−3)	(84	+42)	74° E	23° N
4	Pillow lava	6	4·41	3·2	74	+3	82	−24	99 E	5 S
5	Pillow lava	4	3·36	4·7	64	+27	63	−16	112 E	9 N
6	Pillow lava	6	4·28	2·9	62	+4	66	−40	117 E	4 S
7	Pillow lava	5	4·69	13·0	64	−25	70	−52	120 E	15 S
8	Pillow lava	3	2·60							
9	Pillow lava	3	2·79	9·4	46	−54	111	−67	112 E	20 S
10	Pillow lava	6	4·35	3·0	64	−12	84	−65	122 E	32 S
11	Pillow lava	5	3·75	3·2	2	+55	16	+23	151 E	48 N
12	Pillow lava	6	3·65							
13	Pillow lava	5	3·46							
14	Pillow lava	5	2·59							
15	Pillow lava	6	3·40							
16	Pillow lava	3	2·69	6·4	(134	+37)	(98	+44)	63 E	16 N
17	Breccia	4	3·11	3·4	61	+42	50	+3	117 E	24 N
18	Columnar flow	5	2·84							
19	Columnar flow	4	3·05							
20	Columnar flow	4	2·05							
21	Columnar flow	5	4·08	4·3	(165	−17)	(164	+21)	12 E	25 S
22	Pillow lava	3	2·68	6·2	43	+8	48	−39	132 E	16 S

Overall mean for the nine grouped sites

	R	*k*	*D*	*I*	Palaeomagnetic pole Long.	Lat.	$\partial\psi$	$\partial\psi$
in situ	7·35	4·9	57	+6	111° E	22° N	13	26
simple tilt correction	7·46	5·2	61	−34	119 E	2 N	16	28
two stage tilt correction			58	−34				

Thomas 1956). Sites were distributed through its thickness with numbers 4 to 16 and 22 being in pillowed lava and sites 3 and 18 to 21 being in separate units of columnar flows. Site 17 was taken in a lava breccia between pillowed flows. Total NRM directions are statistically significant at 14 sites and 13 fall in a group; the other six sites are non-significant. AF demagnetization (Fig. 12(c)) was followed by magnetic cleaning of the collection at 600 and 800 Oe (peak), the former being used for a final Fisher analysis (Table 7). Twelve sites have statistically significant stable remanence; nine of these are in pillowed lavas and one in volcanic breccia and nine of these form a single group for which overall statistics are given in Table 7. The anomalous pillow lava of site 16 is omitted from the analysis by the rejection criterion of Watson & Irving (1957). The other two significant sites are in non-pillowed flows and are magnetized in the upper south-east octant, their magnetization remains unexplained. The *in situ* mean magnetization of the nine sites is not matched by the supposed geomagnetic field in Ordovician or any later time in the region (see Paper VI). However after correction for the folding of these beds, which straddle the Llanwnda anticline, the remanence is close to that postulated for Siluro–Devonian time in Britain. Although comparison of precisions in a 'fold test' is not conclusive, a secondary origin for the remanence is more likely because the breccia carries a systematic magnetization. Also the iron oxides in these rocks are mainly by-products of the albitization of the sequence which also partially affected the sills intruding the volcanic rocks.

After tilt correction, the mean is $D = 61°$, $I = -34°$; if the plunge of the Llanwnda anticline and the residual dip of its flanks are treated successively the declination is increased by 7° but inclination is not materially altered. This stable remanence is 'pre-folding' but secondary, and dates from the time of the Caledonian orogeny in this region.

9.3 *Basic intrusive rocks*

Twenty-one sites in intrusive rocks on the north coast of Pembrokeshire outcropping from Aber Gwaun (Fishguard) westwards to Abereiddy Bay (Fig. 11) have been sampled by Almond (1971), Channell (1971) and Rice (1971). Three sites are in keratophyres which are possibly extrusive, and 17 are from sill-like bodies predating the folding and believed to be contemporaneous with the volcanic activity producing the Fishguard Volcanic Series and similar rocks. Site 67 is from a large plug-like mass which is unlikely to have been rotated on the scale of the principal deformation and hence may be approximately in its original orientation.

Only 12 sites have statistically significant NRMs reducing to nine after AF cleaning in fields between 300 and 600 Oe. The demagnetization curves and within-site scatter both suggest poor stability in many cases. *In situ* site means are randomly scattered (Fig. 13(b)). However after simple tilt correction (Fig. 13(a)) five site mean directions are downward to the south and two are upward to the north. These directions are close to the axis of magnetization deduced from the Builth Volcanic Series (Paper 1) of similar age some 120 km to the east. This suggests that an original Ordovician magnetization is carried by these rocks, but because the evidence is poor no mean direction or pole position is quoted.

10. Llandovery lavas in the Tortworth Inlier

Two andesite traps of Upper Llandovery age outcrop around a shallow syncline in the Tortworth inlier, Gloucestershire (Curtis 1972) and have been sampled at five localities (Fig. 14). At sites 1 and 18 total NRM is well grouped (with the exception of one single core at the latter which is rejected). These two sites—one from each of the traps—show a high degree of stability with AF demagnetization. Their mean,

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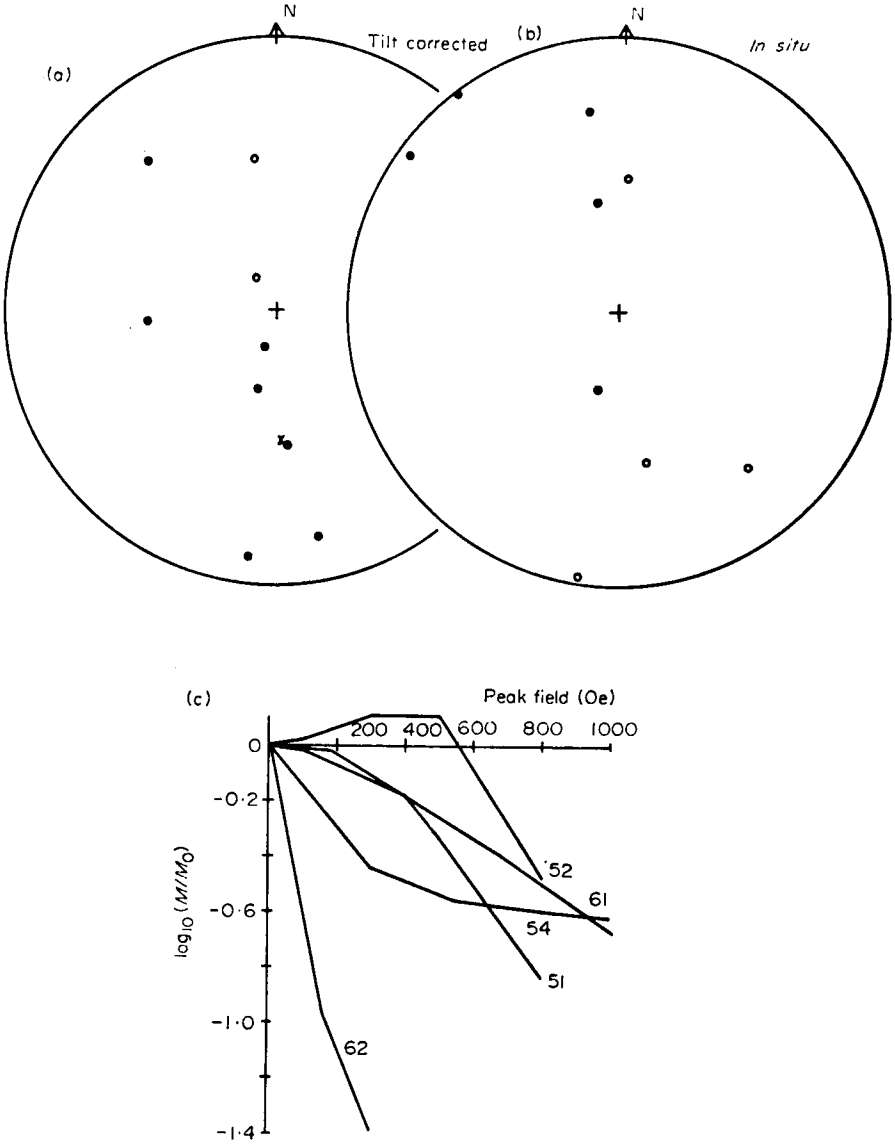


FIG. 13. Basic intrusives of the north coast of Pembrokeshire. Significant site-mean directions after AF cleaning (a) dip corrected as appropriate (see text), and (b) *in situ*, (c) AF demagnetization curves. X is the mean direction in Ordovician rocks in the Builth inlier, for comparison.

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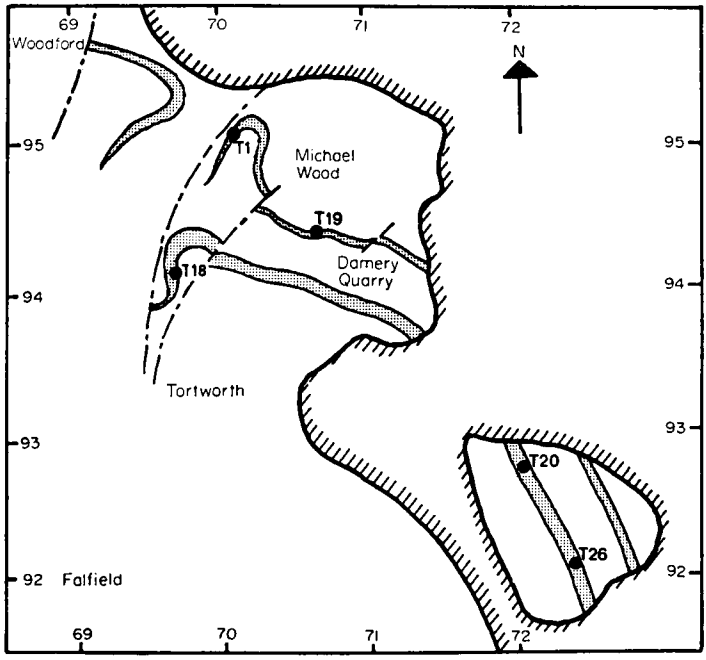


FIG. 14. Tortworth inlier, general geology and sampling sites.

irrespective of polarity is $D = 261°$, $I = +34°$, $k = 12{\cdot}0$ based on eleven samples. Before tilt correction these two site means are 127° apart, after tilt correction they are 147° apart and hence may be regarded as of opposite polarity. Two further sites (20, 26) have total NRMs widely scattered about an ill-defined mean close to that of site 1, but they are unstable to AF treatment and within site dispersion is random after attempts at AF cleaning. At site 19 only two samples were collected and no systematic remanence has been observed.

11. Middle and Upper Devonian Lavas in the Orkneys and Shetlands

Twelve sites from four areas of Middle and Upper Devonian volcanicity in the Orkneys and Shetlands have been sampled, and are dealt with here in probable stratigraphic order.

The major area of Old Red Sandstone outcrop in the Shetland Isles is west of the Walls Boundary Fault where it unconformably overlies metamorphic basement. A considerable thickness of basalts, andesites, rhyolites, agglomerates and tuffs interbedded with the sediments is best developed at Esha Ness and was sampled by us at various stratigraphic levels in the sequence described by Finlay (1930):

7. Upper Tuff
6. Upper Andesite Series (S7)
5. Lower Tuff (S5)
4. Lower Andesite Series (S6)
3. Rhyolite
2. Doleritic basalt (S3, S4)
1. Basement Series of tuffs and tuffaceous sediments.

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Table 8

Dip corrected remanence direction in Middle and Upper Devonian lavas in the Orkneys and Shetlands

Site	Nat. Grid Ref.	Total NRM					Treatment	After cleaning				
		N specimens	*R*	*k*	*D*	*I*		*N*	*R*	*k*	*D*	*I*
Esha Ness, Shetland												
S3	245786	3	2·88	16·2	98	+22		no stable direction				
S4	243784	4	3·89	27·4	269	+29		no stable direction				
S6	240778	8	6·13	3·7	147	+55	170–620° C	1			(125	+60)
S5	242782	5	not significant				400° C	1			216	+28
S7	213770	5	not significant				400° C/600 Oe	2			33	−54
Point of Ayre, Mainland Orkney												
R3	590055	9	8·14	9·4	262	+54	325° C/600 Oe	8	7·58	16·7	273	+67
Hoy, Orkney												
R1	264887	11	10·03	10·3	209	+63	400 Oe	8	7·50	13·9	197	+50
R2	210045	9	8·05	8·4	201	+38	none	9	8·05	8·4	201	+38
R4	196990	8	7·19	8·7	231	+48	400 Oe	7	6·47	11·3	232	+42
R5	187008	6	not significant									
Walls Peninsula, Shetland												
S1	300580	4	not significant				400 Oe	1			28	−41
S2	301580	1			132	+8		no stable direction				
	Mean of seven grouped sites after AF and thermal cleaning:							7	6·67	18·4	216	+46
	Corresponding Palaeomagnetic Pole 327° E 2° N ($\partial\psi$, $\partial\chi$) = (12, 18).											

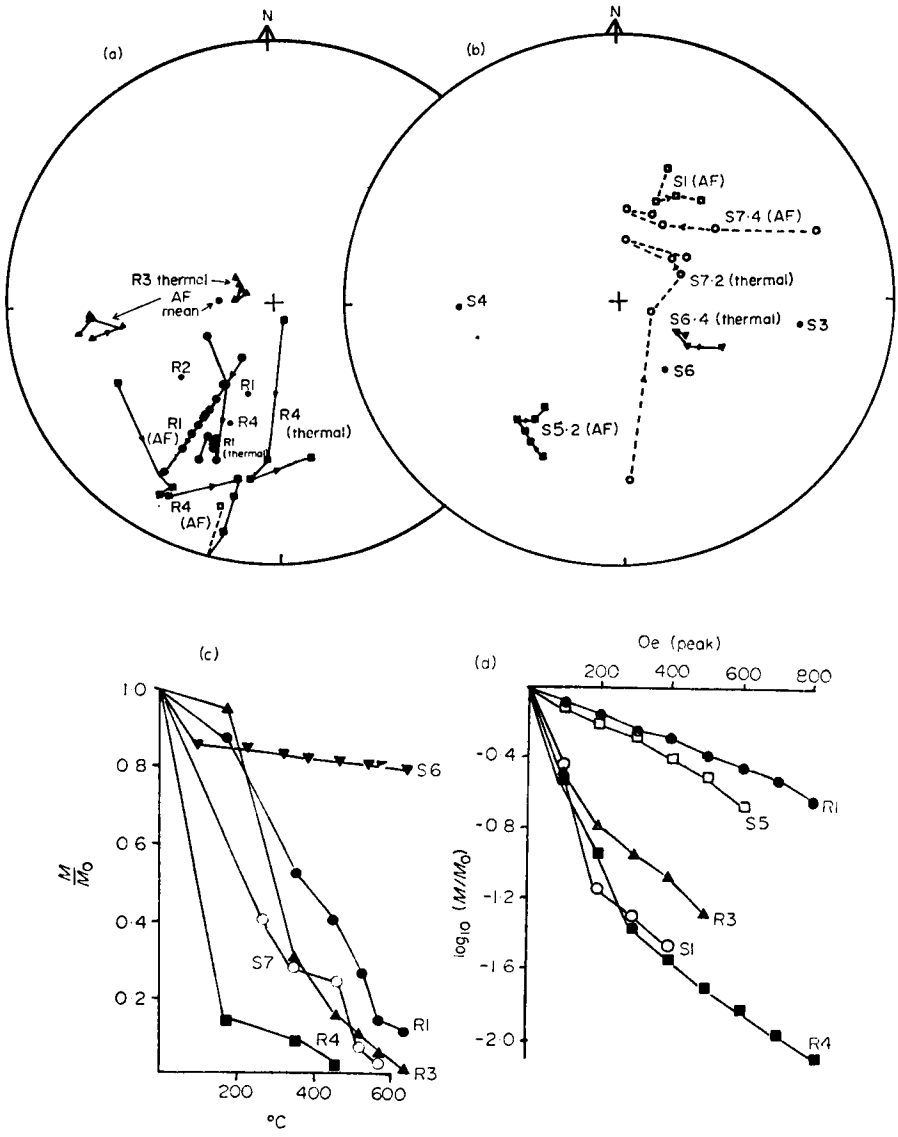


FIG. 15. (a) Orkneys: all statistically significant site means after AF cleaning, and examples of progressive AF and thermal demagnetization. (b) Shetlands: all statistically significant site mean directions of total NRM (small symbols) and all data showing systematic changes in remanence throughout progressive demagnetization (larger symbols). (c) and (d) demagnetization curves for the same specimens.

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Flinn *et al.* (1968) regard the overlying sediments as lower Givetian and estimate an age of *c.* 370 My for the volcanics. The lavas are generally strongly altered leaving augite as the only fresh mineral with secondary chlorite, calcite, chalcedony, agate and iron oxides sometimes in a devitrified matrix. The welded tuffs are rich in phenocrysts of K-feldspar which is often sericitized and replaced by carbonate (Flinn *et al.* 1968). Only three of the five sites show statistically significant grouping of NRM (S3, S4, S6). All have positive inclinations but they do not agree with each other (Table 8). Progressive thermal demagnetization reveals a wide spectrum of critical blocking temperatures (Fig. 15). Only at sites S5 and S6 does more than half the remanence survive heating above 400 °C. Progressive AF demagnetization produces analogous results, the same two sites (S5, S6) exhibiting the greatest stability. Sites S3, S4 show poor stability by both methods. The remanence direction of site S5 reached no stable end-point during thermal treatment but AF demagnetization produces stable remanence in the vicinity of $D = 216°$, $I = +28°$; S6 shows thermal stability near $D = 125°$, $I = +60°$ throughout the temperature range; the high blocking temperature fraction and high coercivity fraction at site S7 agree near $D = 33°$, $I = -54°$.

In Mainland Orkney at the Point of Ayre a single site (R3) in an amygdaloidal basalt, about 4 m thick, occurs within the Eday Flags, part of the local Middle Old Red Sandstone succession. Total NRM is well grouped about $D = 262°$, $I = +54°$. Although partial demagnetization reveals a wide range of both critical blocking temperatures and remanence-coercivity; direction changes throughout progressive treatment are small. Taking maximum SI as the criterion for optimum cleaning, treatment at 520 °C or 600 Oe (peak) is appropriate, and the two procedures both lead to an estimate $D = 273°$, $I = +67°$ ($k = 16{\cdot}7$) for the stable remanence.

The Hoy Volcanics overlie an unconformity at the top of the Middle Old Red Sandstone in the Orkneys, and are regarded as of early Upper Old Red Sandstone age. They consist chiefly of olivine–basalt containing phenocrysts of olivine and feldspar in a groundmass of plagioclase, augite and iron ore (Wilson *et al.* 1935). They were sampled in four areas. A highly vesicular and partly amygdaloidal basaltic lava was collected near Melsetter in southern Hoy (R1). Porphyritic basalt was sampled on the north-eastern slopes of Lounders Fea (R2). A strongly oxidized basaltic flow was collected at the base of the 'Old Man of Hoy' (R5) and porphyritic basalt from Rackwick Bay (R4). The latter two localities were investigated by Storetvedt & Petersen (1972).

At R5 total NRMs are not significantly grouped; elsewhere within-site precision k is at least 8, and the overall *in situ* mean of three sites is well defined at $D = 206°$, $I = +53°$ ($N = 3$, $R = 2{\cdot}95$) *in situ*. Correction for the gentle local dips makes little difference ($D = 213°$, $I = +51°$, $R = 2{\cdot}92$). From detailed magnetic and mineralogical studies Storetvedt & Petersen (1972) recognize the complexity of remanence in these rocks. They find total NRMs with steep inclination and on partial demagnetization observe a southerly movement towards a direction near $D = 205°$, $I = +8°$, which they regard as an ancient remanence from which a viscous component close to the present geomagnetic field has probably been removed. The new results at R4 are in complete agreement with these conclusions but at sites R1 and R2—not visited by Storetvedt & Petersen—there is evidence for a stable remanence with a steeper south-westerly inclination.

At site R1 a southerly change of direction with progressive demagnetization at 100 Oe steps is observed, but between 200 and 600 Oe this variation amounts to only 11°; this is reflected in high values of SI in this range, and indicates a stable direction around $D = 197°$, $I = +50°$. This is confirmed by thermal demagnetization showing that this component corresponds to critical blocking temperatures in the range 350 °C–520 °C. At site R2 the bulk of the remanence is unstable, being removed in 400 Oe, leaving only 1 per cent of the total NRM ($<10^{-6}$ Gauss); one of the two

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pilot specimens does not yield a stable end-direction, but the other shows a systematic component near $D = 60°$, $I = -72°$ corresponding to AF coercivity in the range 400–1000 Oe. Total NRM has been used in the statistical analysis (Table 8).

The samples from the Old Man of Hoy (R5) showed poor stability, but there is some indication in the thermal demagnetization data that the high critical blocking temperature direction is north-westerly and moderately negative. Compared with the axis of shallow south–south-westerly magnetization at Rackwick Bay (i.e. $D \simeq 230°$, $I \simeq +40°$ or its opposite $D \simeq 50°$, $I \simeq -40°$, the (negative) inclination at R5 is steeper and the declination more easterly.

Finally in western Shetland two sites (S1, S2) were sampled from hydrothermally altered basalts of the Clousta Lava Series (Finlay 1930) which occur within the thick sequence of Walls Sandstones. The age of these rocks is not well defined: they could be as old as Middle Devonian but, according to Flinn *et al.* (1968) are most likely to be late Devonian in age, a younger limit being provided by the 335 My age of a granite intruding the sandstone sequence at the southern margin of its outcrop. The only indication of stability in these lavas is at site S1 where the direction is near $D = 28°$, $I = -41°$ during partial demagnetization up to 400 Oe peak. Because there is no obvious stratigraphic variation in the axis of stable magnetization in the Middle and Upper Devonian rocks described here, an overall Fisher analysis of the group is presented in Table 8.

12. Upper Devonian and Lower Carboniferous sediments near Bristol

In the original palaeomagnetic study of the Old Red Sandstone sediments of the Anglo–Welsh cuvette (Creer 1957b) and their thermal demagnetization characteristics (Chamalaun & Creer 1964, Chamalaun 1964) all the rocks investigated were of Lower Old Red Sandstone age with the exception of the Portishead Beds near Bristol which are of Upper Old Red Sandstone age. The results obtained from the Portishead Beds were not distinct from those of the older sediments. Subsequent suggestions of a change in the characteristic Devonian palaeomagnetic field direction in Britain during Devonian times (Creer & Embleton 1967; Evans *et al.* 1971) make further investigation appropriate. Sallomy (1972) reports results from ten sites in the Upper Series of the Portishead Beds (Wallis 1928) on the coast between Portishead and Clevedon where they dip approximately 20° to the SE (sites P1, P2, P5–9) and in the core of the Westbury-on-Trym anticline (Kellaway & Welch 1948) where they dip northwards at about 50° (sites P10–P12). Additionally, sites P3 and P4 were located in a red sandy limestone of Lower Carboniferous age at Portishead Bay.

The difference in dip between the two sampling areas cannot give a conclusive fold-test of stability because on the coast the NRMs lie close to the local strike. Of the three inland sites one is randomly magnetized and another is anomalous in relation to all the others and defies interpretation. The overall mean directions at the other eight sites before and after tilt correction are virtually identical (Table 9). Thermal demagnetization up to 500 °C gives no indication of a multi-component remanence. Because heating was done in air, sharp increases in magnetic moment and large direction changes after heating beyond 500 °C must be ascribed to oxidation. This 'failed' experiment suggests that the original remanence was carried by relatively unoxidized minerals (which are the more likely to be primary in these sediments) and that these were oxidized to high susceptibility phases (? 'haematite') during heating.

The two sites in Carboniferous Limestone (P3 and P4) were from opposite limbs of a small asymmetrical syncline. The difference between their mean NRM directions reduced to 17° after tilt correction, showing that the bulk of the remanence predates the (Hercynian) folding. These 'dip-corrected' directions are in close agreement with the result from the Portishead Beds. Taking all ten significant sites

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Table 9

NRM of Portishead Beds and overlying Carboniferous Limestone

Site	Nat. Grid Ref.	*N* (cores)	*R*	*k*	*D* *in situ*	*I* *in situ*	*D* dip corrected	*I* dip corrected
Portishead Beds								
P1	409720	10	9·38	14·6	205	+38	193	+19
P2	409720	6	5·98	300·0	198	+5	199	−15
P5	460770	10	9·88	73·5	201	+8	197	+8
P6	460770	9	8·63	21·8	220	+25	210	+30
P7	460770	8	7·26	9·5	226	+11	221	+21
P8	460770	9	8·78	36·6	197	+5	195	+5
P9	460770	6	5·78	22·5	182	+11	180	+12
P10	540767	7	6·86	42·0	(116	+52)	(64	+38)
P11	555748	4	not significant					
P12	548743	7	6·65	17·2	196	+13	194	+1
Eight sites (excluding P10)		8 sites	7·67	21·1	203	+15		
			7·65	20·1			198	+10
Lower Carboniferous								
P3	465774	7			195	+22	196	−1
P4	465774	6			191	−41	182	+10
Ten sites (P1–9, 11, 12)		10	9·23	11·7	201	+10	—	—
			Palaeomagnetic pole 333° E, 30° S (8, 15)					
		10	9·60	22·5	—	—	196	+9
			Palaeomagnetic pole 338° E, 32° S (5, 10)					

together (Table 9), the improvement in k from 11·7 to 22·5 achieved by tilt-correction is significant with better than 90 per cent probability by the F-ratio criterion (McElhinny 1964).

13. Conclusions

In Section 2 of Paper VI an apparent polar wander curve is first established on the basis of detailed studies, *without* using any secondary data (e.g. that described in Sections 5–12 of this paper). The Ordovician to Carboniferous poles fall in three groups: (a) Ordovician, (b) Siluro–Devonian, and (c) Carboniferous (Fig. 16). Some of the secondary data of this paper fall into these same groups. The results may record syngenetic or post-genetic remanence, and since the characteristic directions (a), (b) of older rocks are not repeated by the local palaeomagnetic field in younger times, the age of remanence can be estimated by reference to the direction of magnetization. Other results do not fit the polar curve *precisely* (Fig. 16). Their directions of magnetization can be explained either in terms of rotation about a vertical axis, or a westerly geomagnetic declination during part of Silurian times.

The secondary data allow the polar curve to be refined in a number of respects. First, the pole remained in the vicinity of 10° S, 0° E from the early Llanvirnian through to Ashgillian times. Second, the westerly direction (Group (b)) appears

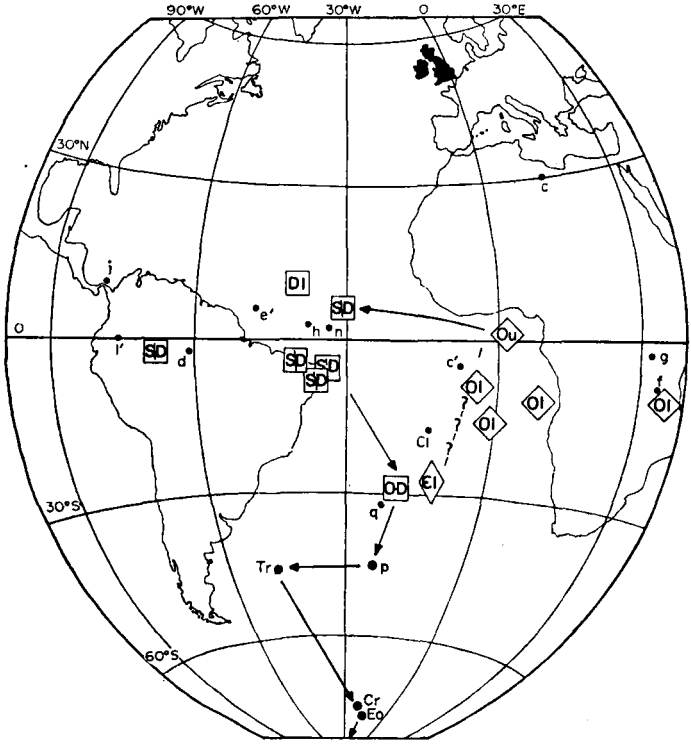


FIG. 16. Apparent south polar wander curve for the British Isles in Phanerozoic times. The pre-Carboniferous path is based on the primary results which are boxed in the diagram. The age notation is Lower Cambrian (Cl), Lower Ordovician (Ol), Upper Ordivician (Ou), Siluro–Devonian (S/D) and Lower Devonian (Dl). Poles derived from second rank studies are indicated by the small dots, letters referring to Table 1 of Paper VI. Lower Carboniferous pole Cl. The large dots are mean post-Carboniferous poles of Permian (P), Triassic (Tr), Cretaceous (Cr) and Eocene (Eo) age.

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to be characteristic of the field from the Upper Llandovery right through to the Middle Devonian, whereas previously it was unknown whether it referred to more than the interval late Silurian to early Devonian. A rapid shift near the Ordovician–Silurian boundary is therefore indicated. Third, the pole appears to have reached its 'Carboniferous' position by late Devonian times and hence a further shift is indicated in the Upper (or possibly Middle) Devonian.

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